Retail Sales Performance & Profitability Analysis

Project Overview

This project focuses on analyzing retail sales data to evaluate business performance and profitability. The aim is to identify key factors affecting sales, profit, and overall business growth.

The analysis is performed using Python for data preprocessing and exploratory analysis, SQL for querying structured data, and Power BI for visualization.

The project answers important business questions such as:

- How sales and profit vary over time
- Which categories and regions perform best
- What is the impact of discount on profit
- Which products are less profitable

This project helps in making data-driven business decisions.

Dataset Summary

The dataset contains **9,994 records and 21 columns**, representing retail sales transactions.

It includes a mix of:

- **Numerical data** such as Sales, Profit, Quantity, and Discount
- **Categorical data** such as Category, Sub-Category, Region, and Customer details
- **Date fields** like Order Date and Ship Date

The dataset is **clean with no missing values** and is well-structured, making it suitable for performing sales, profitability, and trend analysis.

Exploratory Data Analysis (EDA) using Python

Exploratory Data Analysis (EDA) was performed using Python to understand the dataset, identify patterns, and prepare the data for further analysis.

Data Loading and Inspection

The dataset was loaded using pandas, and initial inspection was done using functions like `.head()`, `.info()`, and `.describe()` to understand the structure, data types, and basic statistics.

Initial Data Preview `df.head()`

```
df.head(10)
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	S
0	1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.
1	2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.
2	3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.
3	4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.
4	5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.

`df.info()`

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Row ID          9994 non-null   int64
 1   Order ID        9994 non-null   object
 2   Order Date      9994 non-null   object
 3   Ship Date       9994 non-null   object
 4   Ship Mode       9994 non-null   object
 5   Customer ID     9994 non-null   object
 6   Customer Name   9994 non-null   object
 7   Segment         9994 non-null   object
 8   Country         9994 non-null   object
 9   City            9994 non-null   object
 10  State           9994 non-null   object
 11  Postal Code     9994 non-null   int64
 12  Region          9994 non-null   object
 13  Product ID      9994 non-null   object
 14  Category        9994 non-null   object
 15  Sub-Category    9994 non-null   object
 16  Product Name    9994 non-null   object
 17  Sales           9994 non-null   float64
 18  Quantity        9994 non-null   int64
 19  Discount        9994 non-null   float64
 20  Profit          9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

Data Cleaning

Data cleaning was performed to ensure consistency and accuracy:

- Converted **Order Date** and **Ship Date** into datetime format
- Handled inconsistent date formats
- Checked for missing values

Data Type Conversion and Cleaning

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   datetime64[ns]
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[ns](2), float64(3), int64(3), object(13)
memory usage: 1.6+ MB
```

Feature Engineering

New columns were created to support analysis:

- **Order Year** extracted from Order Date
- **Order Month** extracted from Order Date
- **Profit Margin** calculated using Profit and Sales

Feature Engineering Output

```
df['order_year'] = df['order_date'].dt.year
df['order_month'] = df['order_date'].dt.month
df['profit_margin'] = df['profit']/df['sales']
```

```
df.columns
```

```
Index(['row_id', 'order_id', 'order_date', 'ship_date', 'ship_mode',
       'customer_id', 'customer_name', 'segment', 'country', 'city', 'state',
       'postal_code', 'region', 'product_id', 'category', 'sub-category',
       'product_name', 'sales', 'quantity', 'discount', 'profit', 'order_year',
       'order_month', 'profit_margin'],
      dtype='object')
```

Data Understanding

Basic statistical analysis was performed to understand the distribution of numerical variables such as Sales, Profit, and Quantity.

Statistical Summary `df.describe()`

```
df.describe(include='all')
```

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	city	...	category	sub-category
count	9994.000000	9994	9994	9994	9994	9994	9994	9994	9994	9994	...	9994	9994
unique	NaN	5009	NaN	NaN	4	793	793	3	1	531	...	3	17
top	NaN	CA-2017-100111	NaN	NaN	Standard Class	WB-21850	William Brown	Consumer	United States	New York City	...	Office Supplies	Binders
freq	NaN	14	NaN	NaN	5968	37	37	5191	9994	915	...	6026	1523
mean	4997.500000	NaN	2016-04-30 00:07:12.259355648	2016-05-03 23:06:58.571142912	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
min	1.000000	NaN	2014-01-03 00:00:00	2014-01-07 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
25%	2499.250000	NaN	2015-05-23 00:00:00	2015-05-27 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
50%	4997.500000	NaN	2016-06-26 00:00:00	2016-06-29 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
75%	7495.750000	NaN	2017-05-14 00:00:00	2017-05-18 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
max	9994.000000	NaN	2017-12-30 00:00:00	2018-01-05 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
std	2885.163629	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN

Data Analysis using SQL

Structured Query Language (SQL) was used to analyze the dataset stored in a MySQL database. Various queries were executed to extract meaningful insights related to sales performance and profitability.

Database Setup and Data Retrieval

The dataset was stored in a MySQL database and basic queries were executed to view and verify the data.

Database Creation and Data Preview

```
CREATE DATABASE sales_analytics ;
USE sales_analytics;

select * from sales_data limit 20;
```

Total Sales and Profit

A query was executed to calculate the overall sales and profit of the business.

Total Sales and Profit Query Output

total_sales	total_profit
2297200.86	286397.02

Insight:

- Provides an overview of overall business performance
- Helps understand total revenue and profitability

Sales and Profit Trend Analysis

Time-based analysis was performed to understand how sales and profit vary over time.

Year-wise Analysis:

Year-wise Sales and Profit

order_year	total_sales	total_profit
2014	484247.5	49543.97
2015	470532.51	61618.6
2016	609205.6	81795.17
2017	733215.26	93439.27

Month-wise Analysis:

Month-wise Sales and Profit

order_year	order_month	total_sales	total_profit
2014	1	14236.89	2450.19
2014	2	4519.89	862.31
2014	3	55691.01	498.73
2014	4	28295.34	3488.84
2014	5	23648.29	2738.71
2014	6	34595.13	4976.52
2014	7	33946.39	-841.48
2014	8	27909.47	5318.1
2014	9	81777.35	8328.1
2014	10	31453.39	3448.26
2014	11	78628.72	9292.13
2014	12	69545.62	8983.57
2015	1	18174.08	-3281.01
2015	2	11951.41	2813.85
2015	3	38726.25	9732.1
2015	4	34195.21	4187.5
2015	5	30131.69	4667.87
2015	6	24797.29	3335.56
2015	7	28765.32	3288.65

order_year	order_month	total_sales	total_profit
2015	8	36898.33	5355.81
2015	9	64595.92	8209.16
2015	10	31404.92	2817.37
2015	11	75972.56	12474.79
2015	12	74919.52	8016.97
2016	1	18542.49	2824.82
2016	2	22978.82	5004.58
2016	3	51715.88	3611.97
2016	4	38750.04	2977.81
2016	5	56987.73	8662.15
2016	6	40344.53	4750.38
2016	7	39261.96	4432.88
2016	8	31115.37	2062.07
2016	9	73410.02	9328.66
2016	10	59687.75	16243.14
2016	11	79411.97	4011.41
2016	12	96999.04	17885.31
2017	1	43971.37	7140.44
2017	2	20301.13	1613.87
2017	3	58872.35	14751.89
2017	4	36521.54	933.29
2017	5	44261.11	6342.58
2017	6	52981.73	8223.34
2017	7	45264.42	6952.62
2017	8	63120.89	9040.96
2017	9	87866.65	10991.56
2017	10	77776.92	9275.28
2017	11	118447.83	9690.1
2017	12	83829.32	8483.35

Insight:

- Identifies growth trends over time
- Helps in understanding seasonal patterns

Category and Sub-Category Analysis

Analysis was performed to identify the most and least profitable categories and sub-categories.

Category-wise Profit Analysis

category	total_profit
Technology	145454.95
Office Supplies	122490.8
Furniture	18451.27

Sub-Category-wise Profit Analysis

sub-category	total_profit
Copiers	55617.82
Phones	44515.73
Accessories	41936.64
Paper	34053.57
Binders	30221.76
Chairs	26590.17
Storage	21278.83
Appliances	18138.01
Furnishings	13059.14
Envelopes	6964.18
Art	6527.79
Labels	5546.25
Machines	3384.76
Fasteners	949.52
Supplies	-1189.1
Bookcases	-3472.56
Tables	-17725.48

Insight:

- Some categories generate higher profit compared to others
- Certain sub-categories may contribute to losses

Region-wise Profit Analysis

A query was used to analyze profit distribution across different regions.

Region-wise Profit Analysis

region	total_profit
West	108418.45
East	91522.78
South	46749.43
Central	39706.36

Insight:

- Identifies high-performing and low-performing regions
- Helps in region-based decision making

Discount Impact Analysis

The relationship between discount and profit was analyzed using SQL queries.

Discount vs Profit Analysis

discount	avg_profit	total_profit
0	66.9	320987.6
0.1	96.06	9029.18
0.15	27.29	1418.99
0.2	24.7	90337.31
0.3	-45.68	-10369.28
0.32	-88.56	-2391.14
0.4	-111.93	-23057.05
0.45	-226.65	-2493.11
0.5	-310.7	-20506.43
0.6	-43.08	-5944.66
0.7	-95.87	-40075.36
0.8	-101.8	-30539.04

Insight:

- Higher discounts tend to reduce profit
- Helps in optimizing discount strategies

Product Performance Analysis

Products with high sales but low or negative profit were identified.

High Sales but Low Profit Products

product_name	total_sales	total_profit
Cisco TelePresence System EX90 Videoconferen...	22638.48	-1811.08
HON 5400 Series Task Chairs for Big and Tall	21870.58	0
GBC DocuBind P400 Electric Binding System	17965.07	-1878.17
High Speed Automatic Electric Letter Opener	17030.31	-262
Lexmark MX611dhe Monochrome Laser Printer	16829.9	-4589.97
Martin Yale Chadless Opener Electric Letter Ope...	16656.2	-1299.18
Riverside Palais Royal Lawyers Bookcase, Royal...	15610.97	-669.54
Bretford Rectangular Conference Table Tops	12995.29	-327.23
Cubify CubeX 3D Printer Double Head Print	11099.96	-8879.97

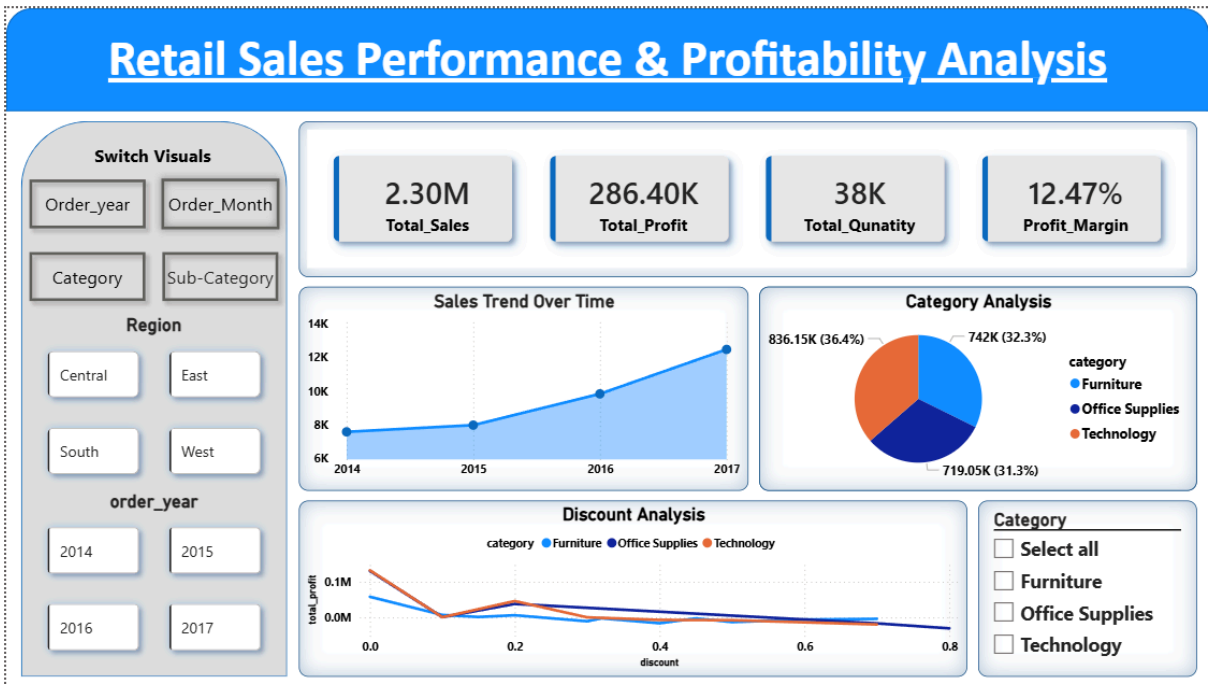
Insight:

- Some products generate high revenue but are not profitable
- Helps in identifying loss-making products

Dashboard in Power BI

An interactive dashboard was created using Power BI to visualize the insights obtained from Python and SQL analysis. The dashboard provides a clear and user-friendly representation of key business metrics and trends.

Power BI Dashboard



Key Features of the Dashboard

The dashboard includes the following components:

Key Performance Indicators (KPIs)

- Total Sales
- Total Profit
- Total Quantity
- Profit Margin

These KPIs provide a quick overview of the overall business performance.

Sales Trend Analysis

A line chart is used to show sales trends over time.

- Toggle between **Year-wise** and **Month-wise** views using bookmarks
- Helps in identifying growth patterns and seasonal trends

Category and Sub-Category Analysis

- Donut chart used for **Category distribution**
- Bar chart used for **Sub-Category profit analysis**
- Bookmark buttons allow switching between category and sub-category views

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Discount Impact Analysis

- A line chart is used to analyze the relationship between **discount and profit**
- Helps understand how increasing discounts affect profitability

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Regional Analysis

- Visual representation of profit across different regions
- Helps identify high-performing and low-performing areas

Interactive Features

- Slicers for **Region** and **Order Year**
- Buttons for switching between different views
- Dynamic filtering across all visuals

Dashboard Design Approach

- Clean and simple layout to avoid clutter
- Focus on key insights rather than too many visuals
- Use of bookmarks and buttons to improve user interaction
- Proper alignment and spacing for better readability

Insights from Dashboard

- Sales show a consistent growth trend over time
- Profitability varies across categories and regions

- High discounts lead to lower profit
- Some sub-categories perform better than others

Business Recommendations

Based on the analysis performed using Python, SQL, and Power BI, the following recommendations are suggested to improve business performance and profitability:

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1. Optimize Discount Strategy

High discounts have a negative impact on profit.
→ The company should reduce excessive discounting and apply discounts more strategically.

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2. Focus on High-Profit Categories

Certain categories generate higher profit compared to others.
→ Increase focus on high-performing categories to maximize profitability.

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3. Improve Loss-Making Products

Some products generate high sales but low or negative profit.
→ Review pricing, cost, and discount strategy for these products.

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4. Strengthen High-Performing Regions

Some regions contribute more to overall profit.
→ Expand business operations and marketing efforts in these regions.

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5. Monitor Low-Performing Areas

Regions or categories with low profit need attention.
→ Identify reasons such as high cost or low demand and take corrective actions.

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6. Improve Profit Efficiency

Sales are increasing, but profit growth is not proportional.
→ Focus on cost control and improving profit margins.

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7. Data-Driven Decision Making

Use dashboards and analytics regularly to monitor performance.
→ Helps in making informed business decisions.

Conclusion

This project successfully analyzed retail sales data to understand business performance, profitability, and key factors affecting revenue.

Using Python, the data was cleaned and explored to identify patterns and trends. SQL was used to perform structured analysis and answer important business questions. Finally, Power BI was used to create an interactive dashboard for clear and effective visualization.

The analysis revealed that sales are increasing over time, but profitability is influenced by factors such as discount, category, and region. It also highlighted that some products generate high sales but are not profitable, and higher discounts tend to reduce overall profit.

The dashboard provides a comprehensive view of the business and allows users to interact with the data for better understanding.

Overall, this project demonstrates how data analysis techniques can be used to extract meaningful insights and support data-driven decision-making in a retail business environment.